Strategy Roadmap

Quant Research Platform · recommendations as of November 4, 2026

> **Recommendation in brief**
>
> 1. **Add risk-based allocation first** (inverse volatility, then risk parity, minimum variance and HRP). It is the largest gap, needs no new data, and gives every alpha idea a stronger benchmark than equal weight.
> 2. **Then time-series momentum / trend following**, sized with inverse volatility. It diversifies everything else on the platform and runs on the ETF universes you already have.
> 3. **Hold off on value, quality, carry and event-driven strategies** until there is a point-in-time fundamentals or futures data source. yfinance cannot support them without look-ahead bias.

What the platform already has

AREA	IMPLEMENTED
Allocation baselines	Equal weight (`ewp_strategy`), fixed weight (`fwp_strategy`)
Optimizer	CVXPY mean-variance with position, turnover, sector, asset-class, beta, volatility and tax constraints
Return signals	Cross-sectional momentum, mean reversion (rank reversal and Bollinger z-score), moving averages, risk/return
Risk inputs	Ledoit-Wolf covariance; realized, EWMA and GARCH volatility forecasts; beta vectors
Model-based	Black-Litterman overlay; Ridge cross-sectional ML predictor
Relative value	Pairs trading (cointegration, z-score entry/exit/stop)
Volatility	VIX basis (term-structure ratio, short/long vol with hysteresis)
Overlays	Volatility targeting, leverage with financing spread, tax-lot accounting

Recommended build order

1. **Inverse volatility.** One signal class and one strategy class, with `VixBasisStrategy` as the template. A good first one to write by hand.
2. **Trend following**, reusing the inverse-volatility sizing: trend signal × 1/σ is the standard construction.
3. **Risk parity and HRP**, added as standing benchmarks so every new idea is compared against something stronger than equal weight.

4. **GTAA, dual momentum and 60/40** as a benchmark set for the ETF universes.
5. **Low volatility and the overlays** once the core set is in place.

Recommended additions, in priority order

Risk-based allocation

PRIORITY 1

Allocations that use only the covariance matrix, not expected returns, which are the noisiest input to any optimizer.

- **Inverse volatility**: weight proportional to 1/σ. A few lines on top of `VolatilityForecastingSignals`.
- **Risk parity / equal risk contribution**: each asset contributes the same share of portfolio risk. A small convex problem that fits the existing CVXPY setup.
- **Minimum variance**: the current optimizer with the return term removed; possibly just a zero-expected-returns option.
- **Hierarchical risk parity (HRP)**: clusters correlated assets before allocating; robust when the covariance estimate is noisy, as it is with 100+ stock universes.

Why	Standard baselines on any research platform, and the honest benchmark: an alpha signal that cannot beat inverse volatility on the same universe is not adding much.
Data	Prices only. Works today.
Fits into	A new signal + strategy pair, modelled on `VixBasisSignal` / `VixBasisStrategy`; minimum variance may be an optimizer flag instead.
Effort	Inverse vol: small. Risk parity and min variance: moderate. HRP: moderate (clustering + recursive bisection).

Time-series momentum / trend following

PRIORITY 2

Each asset is held long when its own trailing return is positive and short (or in cash) when negative, sized by inverse volatility. Distinct from the cross-sectional momentum already on the platform, which ranks assets against each other.

- Common variants: sign of the 12-month return; a blend of 1-, 3- and 12-month lookbacks; price above its moving average.

Why	The core of managed-futures returns and one of the few strategies that performed in both 2008 and 2022. It diversifies the equity-heavy strategies already here.
Data	Prices only; the existing ETF universes are a good fit.
Fits into	Trend signal × inverse-volatility sizing, so it reuses the Priority 1 work directly.
Effort	Small to moderate once inverse volatility exists.

Tactical asset allocation benchmarks

PRIORITY 3

- **Faber GTAA**: hold each asset class only while it is above its 10-month moving average, otherwise cash.
- **Dual momentum (Antonacci)**: choose between US and international equity on relative momentum; move to bonds when absolute momentum is negative.
- **60/40 with periodic rebalancing**: already possible with `fwp_strategy`; make it a standard benchmark next to SPY.

Why	Simple, widely published rules that make credible reference points for any multi-asset idea.
Data	Prices only.
Effort	Small; GTAA is close to `moving_average` plus a holding rule.

Price-only equity factors

PRIORITY 4

- **Low volatility / betting against beta**: long the low-beta or low-volatility names. `Signals.beta_vector` already exists. One of the most consistently documented factor effects.
- **Short-term reversal**: already implemented as rank reversal.

Data	Prices only.
Effort	Small: a new expected-returns signal for `systematic_strategy`.

Overlays applicable to any strategy

PRIORITY 5

- **Drawdown control / CPPI**: reduce exposure after a drawdown threshold; sits beside the existing volatility targeting in `base_strategy`.
- **Regime filter**: cut risk when VIX is above its median or SPY is below its 200-day average. `FeatureBuilder` already computes a high-volatility flag.

Effort	Small each; most of the plumbing exists.

Not yet: blocked on data

STRATEGY	WHAT IS MISSING
Value, quality	Point-in-time fundamentals. yfinance `.info` is a current snapshot, so backtesting on it builds in look-ahead bias.
Carry (FX, bonds, commodities)	Futures or forward curves; the same data gap that limits the VIX basis strategy to ETF proxies.
Event-driven (post-earnings drift, index rebalances)	Event datasets with announcement timestamps.

STRATEGY	WHAT IS MISSING
Intraday / high-frequency	Out of scope: the engine is built around daily bars, which is the right design for this platform.

Fix before comparing strategies

Three issues found while reviewing existing strategies distort side-by-side results today:

> **Pairs trading holds no cash when flat.** `_run_pairs_strategy` zeroes the CASH weight, so idle capital earns nothing instead of the risk-free rate. This alone explains the condensed config's Sharpe of −22. Set CASH to `1 - net exposure`, after checking `Portfolio` does not already credit short proceeds.

> **The mean-reversion strategy drains its cash.** CASH is included in its Bollinger z-scores and always scores about +1.6, so every rebalance tilts it down until it reaches zero. Settings such as `bollinger_window` are also ignored there.

> **Rank reversal trades about 21× a year.** At 30 bps that is roughly 6% a year in costs (2005–2026 backtest: Sharpe 0.60 vs 0.67 for the Bollinger variant at 2.9× turnover). Try a lower `turnover_limit`, less frequent rebalancing, or a longer window.

References

1. Moskowitz, Ooi & Pedersen (2012). Time Series Momentum. *Journal of Financial Economics*.
2. Maillard, Roncalli & Teïletche (2010). The Properties of Equally Weighted Risk Contribution Portfolios. *Journal of Portfolio Management*.
3. López de Prado (2016). Building Diversified Portfolios that Outperform Out of Sample. *Journal of Portfolio Management*.
4. Faber (2007). A Quantitative Approach to Tactical Asset Allocation. *Journal of Wealth Management*.
5. Antonacci (2014). *Dual Momentum Investing*. McGraw-Hill.
6. Frazzini & Pedersen (2014). Betting Against Beta. *Journal of Financial Economics*.

Backtest figures come from runs of the platform's own configs during this review and are not tuned. Effort estimates assume the existing signal → strategy → engine structure.